

The Knowledge Engineer's Crucible

15 Diagnostic Challenges in Propositional Logic & Logic Programming

COM759 – Week 3
Masterclass Review

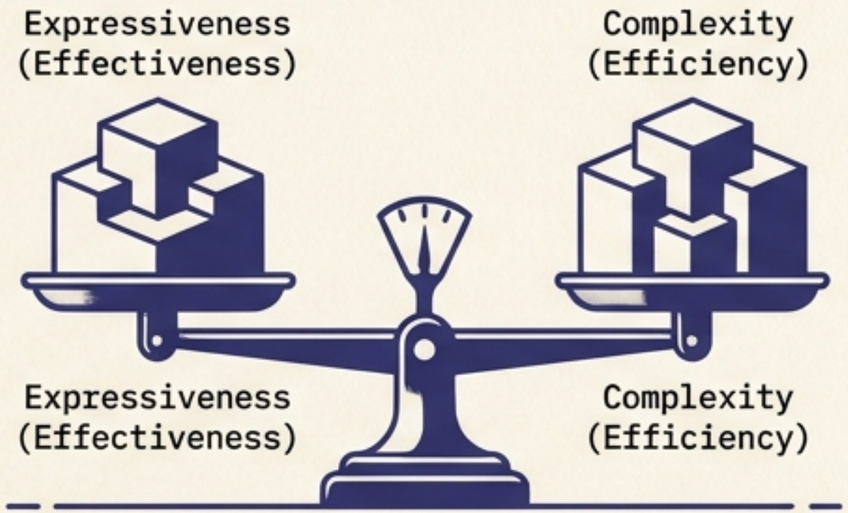
System Calibration Initiated. Before we program an AI to reason, we must **prove we know how to reason ourselves.**

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The Knowledge Engineer's Drafting Table

Q1: According to formal logic modelling, which trade-off defines the relationship between a language's expressiveness and its complexity?

A) Syntax vs. Semantics
B) Effectiveness vs. Efficiency
C) Soundness vs. Completeness



Q2: How is the sentence 'Paola is happy only if she paints a picture' correctly formalised, where p = 'Paola is happy' and q = 'Paola paints a picture'?

A) $q \rightarrow p$
B) $p \leftrightarrow q$
C) $p \rightarrow q$

p only if q

$\neg(p \wedge \neg q) \longrightarrow p \rightarrow q$

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Layer 2: The Enigma of Material Implication

Q3: In the evaluation of the material implication P → Q, if P (the antecedent) is evaluated as False, what is the resulting truth valuend why?

A) False, because the premise has failed.

B) True, because the logical contract was not broken.

C) Undefined, because no causal connection is implied.

D) False, unless Q is also False.



If the premise is false, the implication evaluates to True by default. The logical contract is only broken if you fulfil your end (P is True) but the reward is withheld (Q is False).

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Layer 2: Semantics & Satisfiability

Q4: Consider the formula: $(P \land Q) \rightarrow P$. Which of the following semantic categories best describes this formula?

A)	Satisfiable but not Valid	B)	Unsatisfiable
C)	**Valid (Tautology)**	D)	Contingent

P	Q		$(P \land Q)$	$(P \land Q) \rightarrow P$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	F	F	T

A formula is valid if every interpretation satisfies it.

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Layer 3: Logical Equivalence & Data Representation

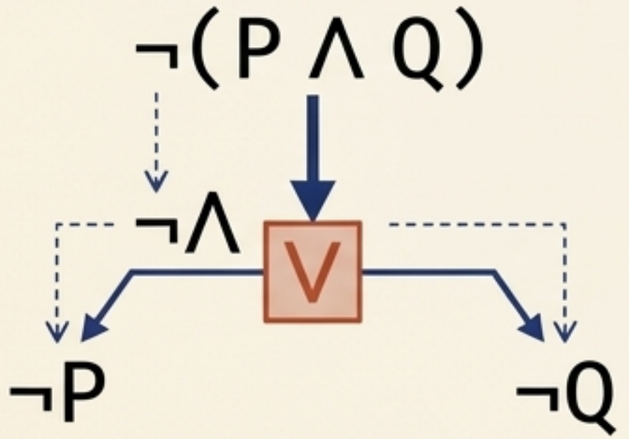
Q5: De Morgan's Laws

When converting a formula to Conjunctive Normal Form (CNF), what is the correct application of De Morgan's laws to the formula ¬(P ∧ Q)?

A) ¬P ∧ ¬Q

B) ¬P ∨ ¬Q

C) P ∨ Q



Visualizing **De Morgan's**: The negation distributes, flipping the logical operator.

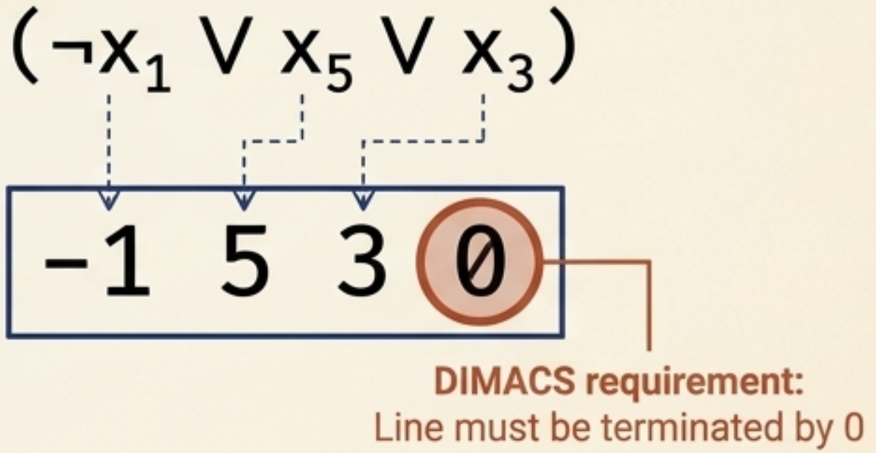
Q6: DIMACS CNF Format

In the **DIMACS CNF** format used by MiniSat, how is the clause $(\neg x_1 \vee x_5 \vee x_3)$ correctly represented?

A) 1 −5 −3 0

B) −x1 x5 x3 0

C) −1 5 3 0



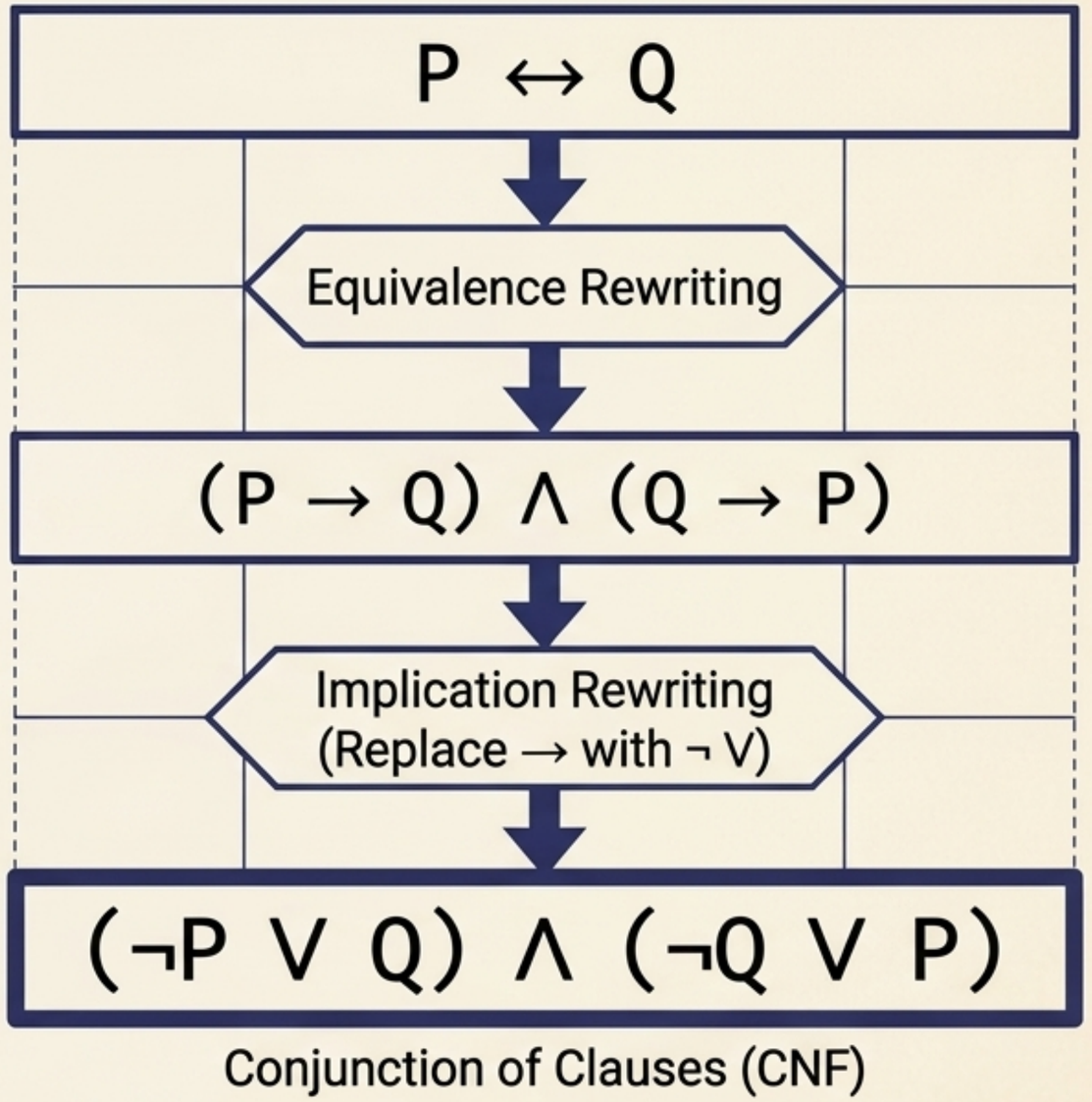
Variables map to signed integers; 0 indicates end of clause.

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Layer 3: Conjunctive Normal Form Pipeline

Q7: Which of the following represents the correct sequence of operations to convert the equivalence formula $P \leftrightarrow Q$ into CNF?

Answer: Rewrite as $(P \rightarrow Q) \land (Q \rightarrow P)$, then as $(\neg P \lor Q) \land (\neg Q \lor P)$



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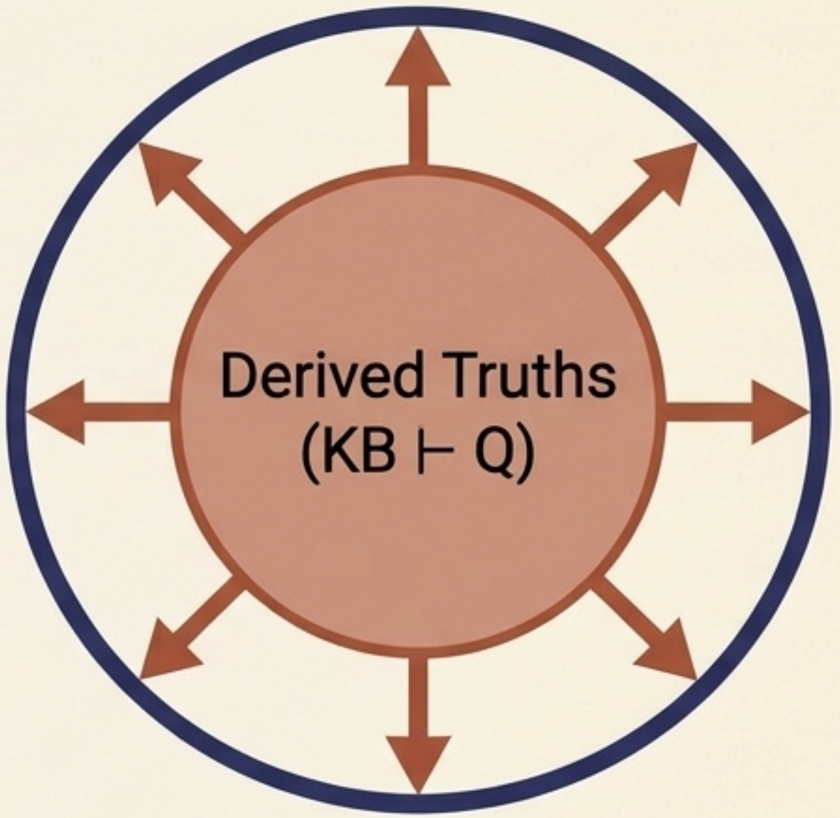
Layer 4: The Mechanics of Proof

Q8: In formal proof theory, an inference rule is described as "Complete" if:

A) Its conclusion is true whenever the premise is true.

B) It does not create any contradictions.

C) It is able to produce every expression that logically follows from the Knowledge Base.



Completeness means the derived universe expands to fully cover and match the entire entailed universe.

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Layer 4: Inference Rules in Action

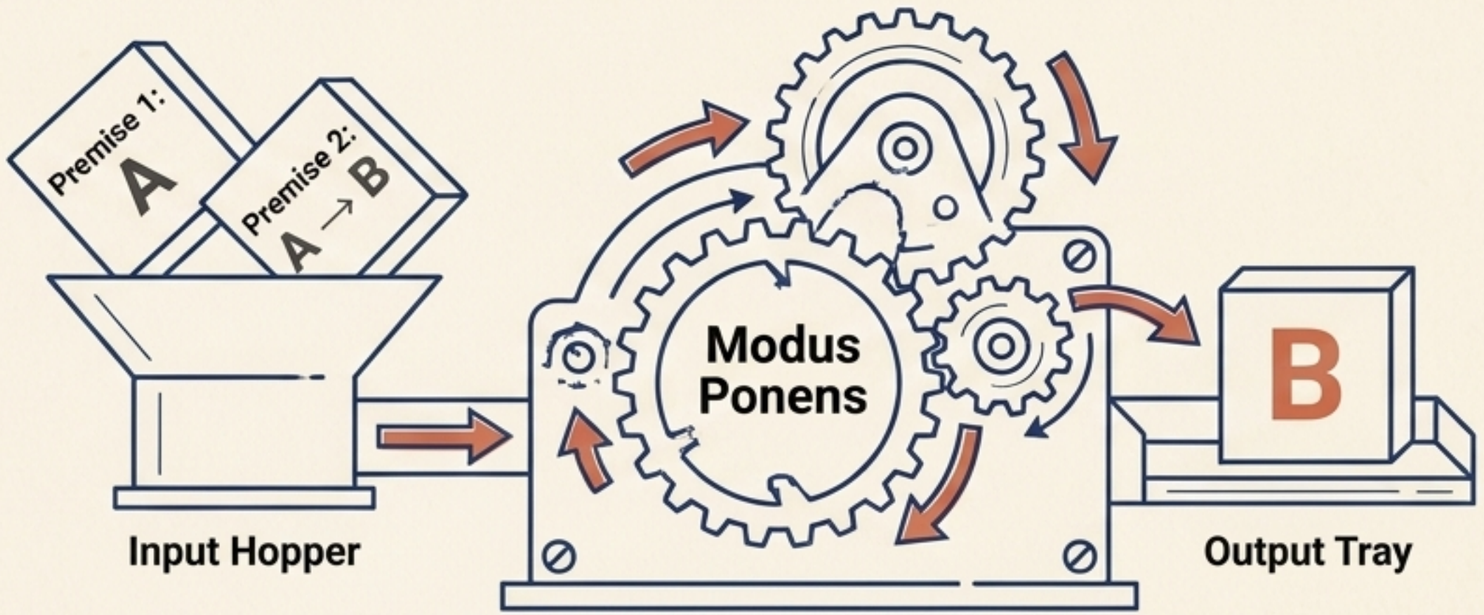
Q9: Given a Knowledge Base containing the premises {A, A → B}, applying the Modus Ponens rule of inference allows us to safely derive which conclusion?

A) A ∧ B

B) ¬A

C) B

D) A ∨ B



If ‘A’ is true, and 'If A then B’ is true, ‘B’ must necessarily be true.

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Layer 5: The Resolution Rule

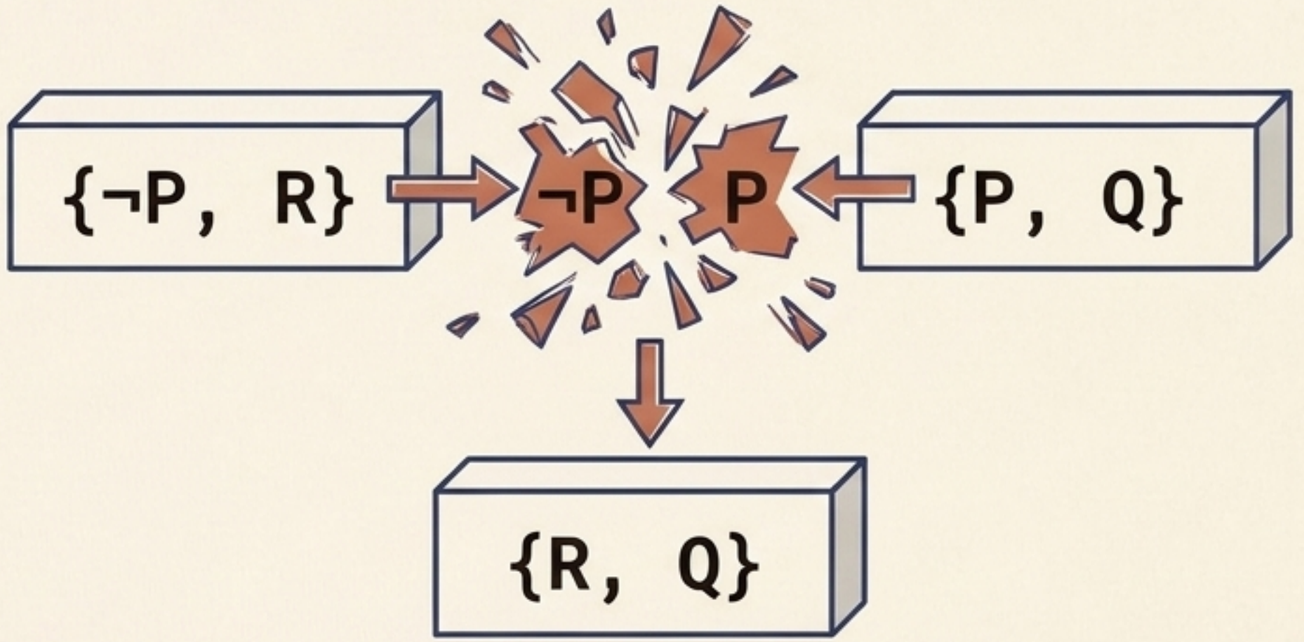
Q10: Apply the Resolution inference rule to the following two clauses: {¬P, R} and {P, Q}. What is the resulting resolvent?

A) {R, Q}

B) {¬P, P, R, Q}

C) {R}

D) The empty clause (□)



Complementary literals annihilate. The surviving literals disjoin.

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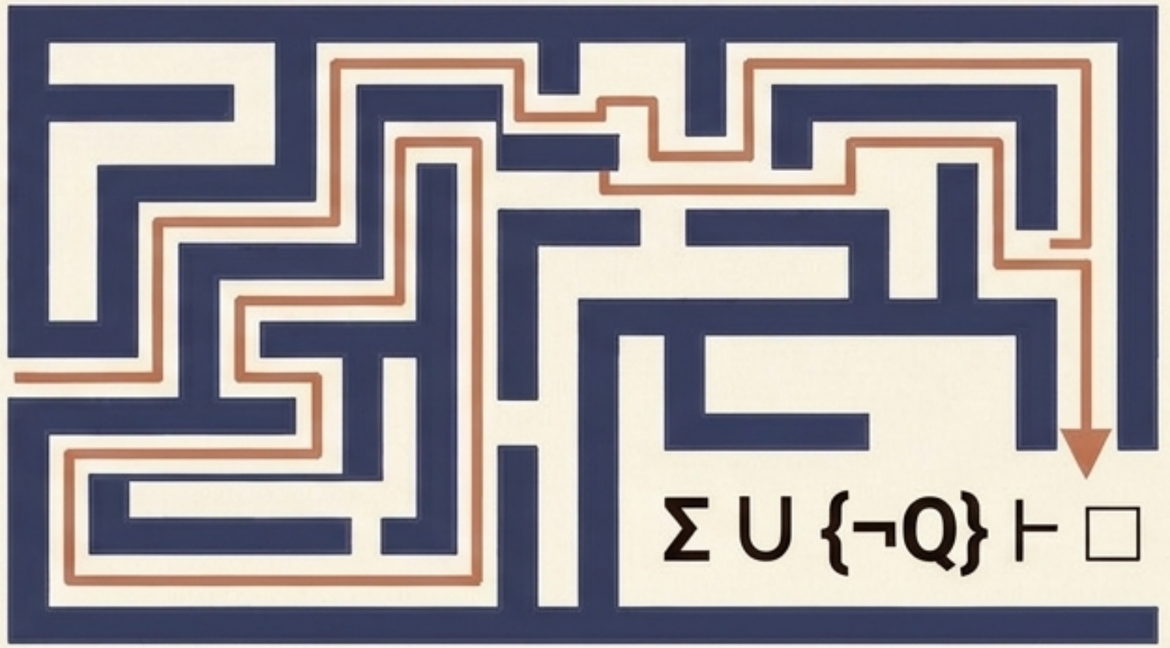
Layer 5: Proof by Contradiction

Q11: When using Resolution for automated reasoning, how do we conclusively prove that a Knowledge Base (Σ) entails a query (Q)?

A) By converting Σ ∪ {Q} to DNF and finding a valid model.

B) By showing that the CNF of Σ ∪ {¬Q} is unsatisfiable.

C) By applying Modus Ponens until Q is derived.



Because resolution is refutation complete, we assume the opposite of our goal (¬Q). If we deduce the empty clause (□), we have found a contradiction, proving the original query Q must be true.

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Layer 5: Horn Clauses & Prolog Syntax

Question 12

Q12: Prolog is built upon a specific subset of predicate logic. What is the defining characteristic of a Horn clause?

- ☑ A) It contains at most one positive literal.
- ☐ B) It is a conjunction of only negative literals.
- ● C) It is an expression in Disjunctive Normal Form.

$$\neg P_1 \vee \neg P_2 \vee Q$$

Single Positive Literal (Head)

Question 13

Q13: In Prolog syntax, how is the logical rule 'John likes X if X is a female and X likes John' correctly written?

A) likes(john, X) -> female(X) AND likes(X, john).

B) likes(john, X) :- female(X), likes(X, john).

$$H \leftarrow B1, B2$$

```
H :- B1, B2.
```

‘:-’ represents implication

‘,’ represents conjunction

Variables capitalized

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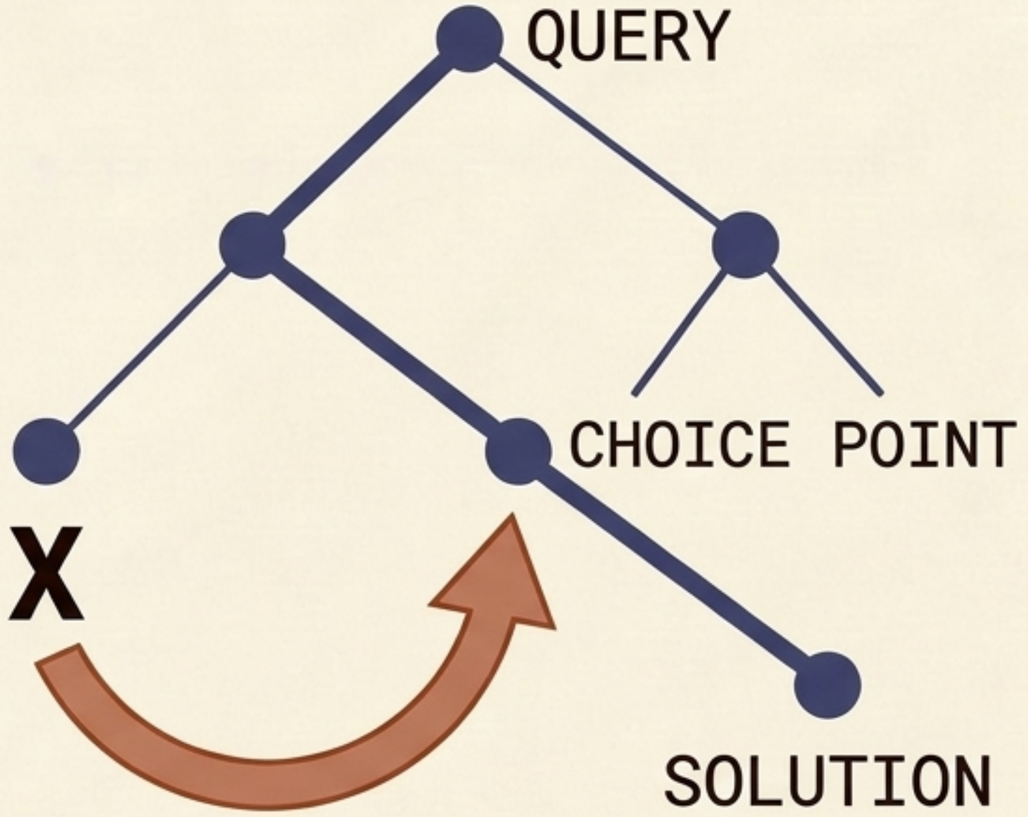
Layer 6: Prolog Execution

Q14: Which two mechanisms drive Prolog's search and inference processes to solve relations and answer queries?

A) Forward chaining and breadth-first search

B) Backtracking and unification

C) Truth table enumeration and valid interpretation



If a path fails, Prolog uses backtracking to return to the previous choice point and tries alternative unifications.

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Synthesis: The Knowledge Matrix (Q15)

Which combination correctly fills the missing parameters for Propositional Logic vs. Prolog?

Answer: Propositional deals only with truth values of complete statements. Prolog uses **Inference by Resolution** on **Horn Clauses** via **Backtracking.**

System	Expressiveness	Inference Method
Propositional Logic	Abstract; deals only with truth values of complete statements	Truth Tables / CNF
First-Order Logic (FOL)	Deals with internal structures, objects, and quantifiers (all, some, no)	General Resolution
Prolog	Solves relations/predicates (FOL subset)	Resolution on Horn Clauses via Backtracking

Final Diagnostic Complete. Screenshot Recommended.

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The Knowledge Engineer's Next Horizon

Next Phase (Week 4): First-Order Logic

Propositional logic is abstract, but AI requires reasoning about internal structures, properties, and quantifiers.

School Research Seminar

Talk: Towards Safe Abductive Learning with Weak Label and Weak Rule
Guest: Prof Yu-Feng Li (School of Artificial Intelligence, Nanjing University)
Date: Wed, 16 October 2024, 1:30pm (UK Time)

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System Calibrated. Crucible Survived.

- ✓ Syntax: Mastered
- ✓ CNF Transformation: Optimised
- ✓ Resolution Proving: Automated
- ✓ Prolog Execution: Compiled

You are no longer just writing code; you are engineering reasoning itself.
Prepare for First-Order Logic. Session terminated.

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